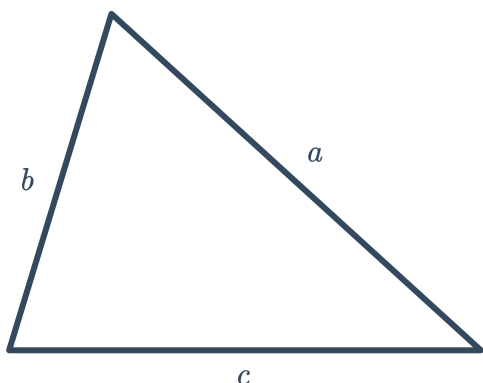


Sine rule

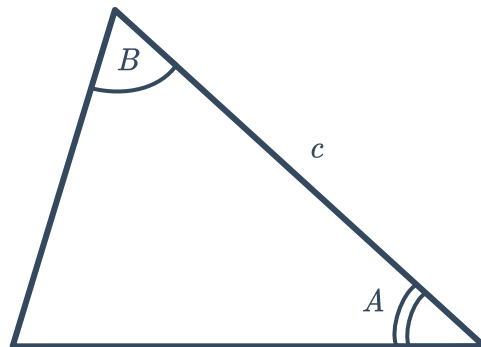


- 1 For each of the given triangles, determine if there is enough information to find all the remaining sides and angles in the triangle using only the sine rule:

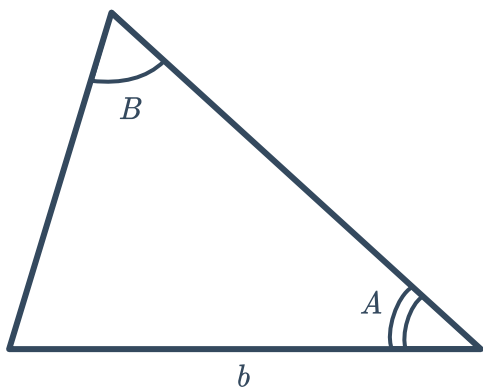
a Three sides are known:



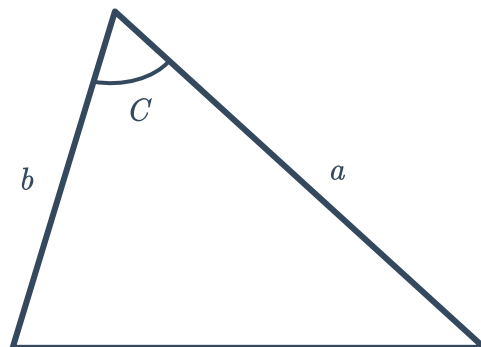
b Two of the angles and the side included between them are known:



c Two of the angles and a side not included between them are known:

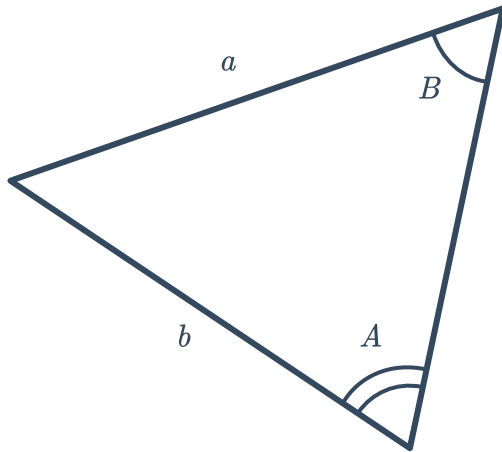


d Two of the sides and an angle included between them are known:

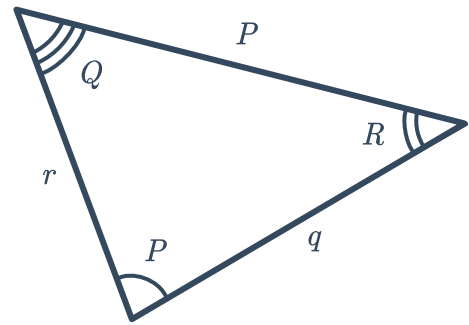


- 2 For each of the following triangles, write an equation relating the sides and angles using the sine rule:

a

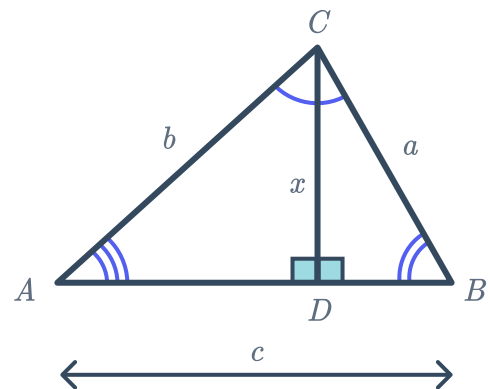


b



- 3 Consider the following diagram:

- Find an expression for $\sin A$ in $\triangle ACD$.
- Find an expression for $\sin B$ in $\triangle BDC$. Then make x the subject of the equation.
- Substitute your expression for x into the equation from part (a), and rearrange the equation to form the sine rule.

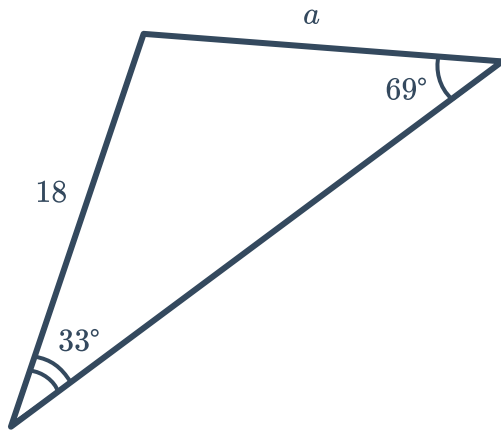




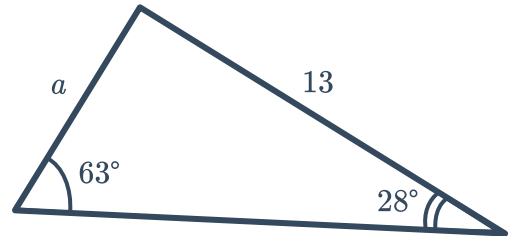
Unknown sides

- 4 For each of the following triangles, find the side length a using the sine rule. Round your answers to two decimal places.

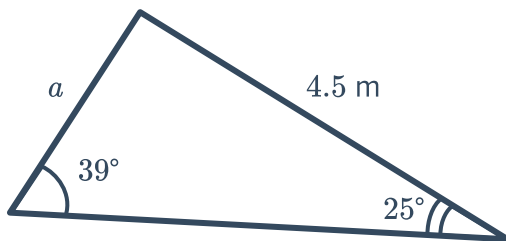
a



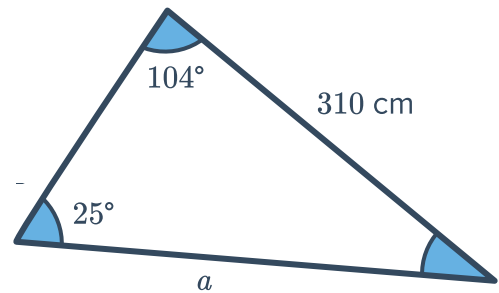
b



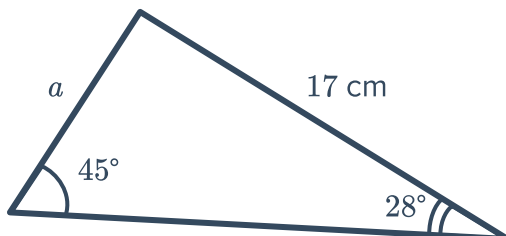
c



d

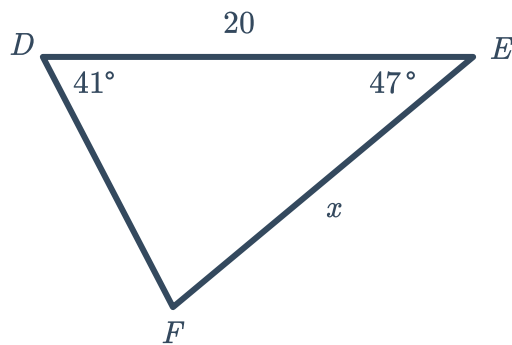


e

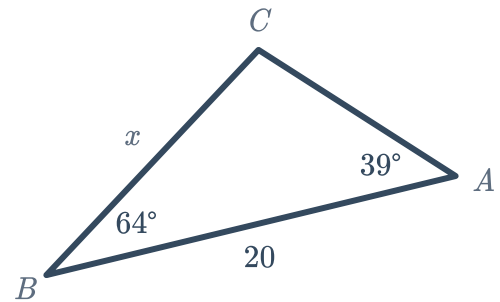


5 For each of the following triangles, find the length of side x , correct to one decimal place:

a



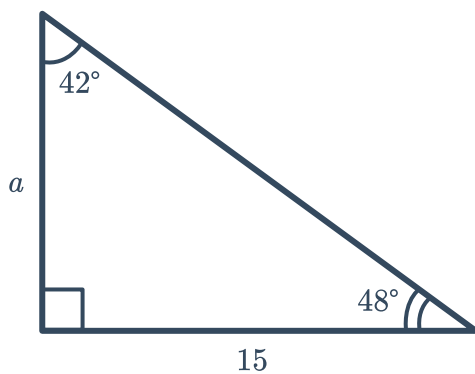
b



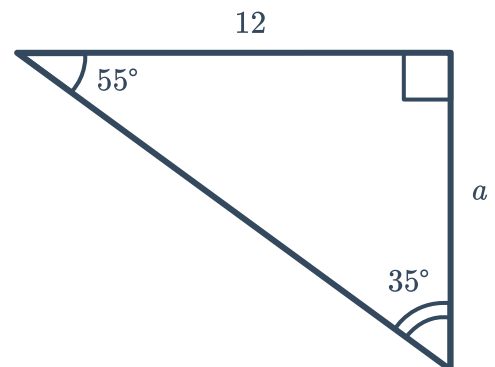
6 For each of the following triangles:

- Find the value of a using the sine rule. Round your answer to two decimal places.
- Use another trigonometric ratio and the fact that the triangle is right-angled to calculate and confirm the value of a . Round your answer to two decimal places.

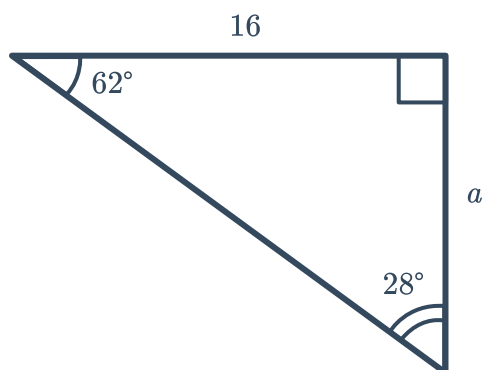
a



b



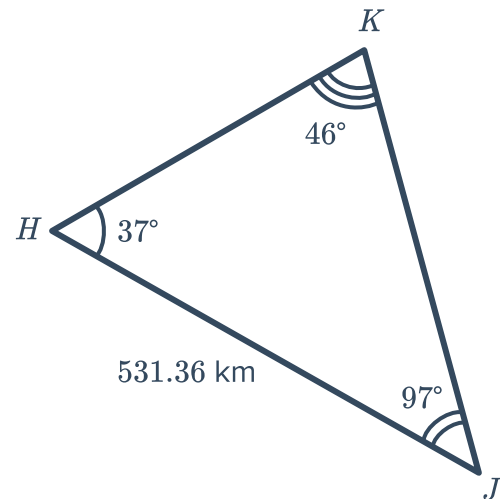
c





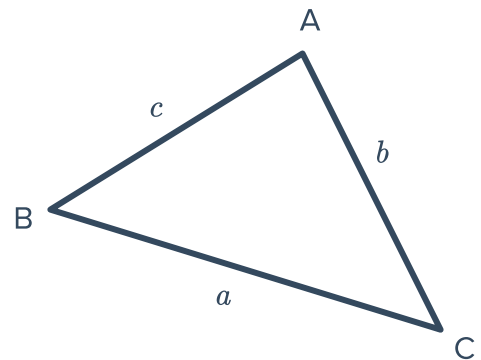
7 Consider the following triangle:

- a Find the length of side HK to two decimal places.
- b Find the length of side KJ to two decimal places.



8 Consider the triangle with $\angle C = 72.53^\circ$ and $\angle B = 31.69^\circ$, and one side length $a = 5.816 \text{ m}$.

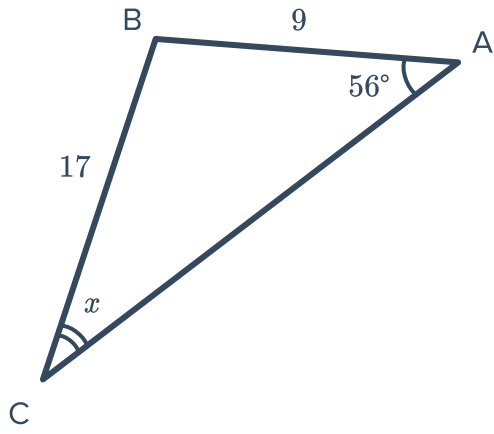
- a Find $\angle A$. Round your answer to two decimal places.
- b Find the length of side b . Round your answer to three decimal places.
- c Find the length of side c . Round your answer to three decimal places.



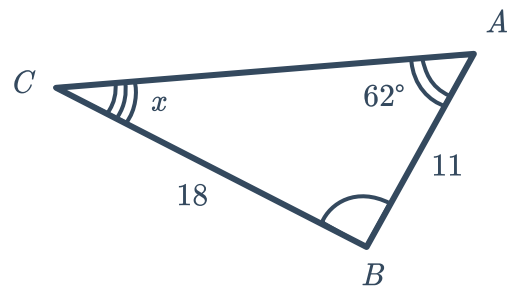
Unknown angles

9 For each of the following diagrams, find the value of the angle x using the sine rule. Round your answers to one decimal place.

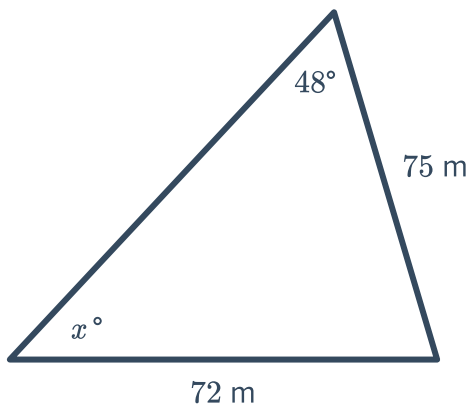
a



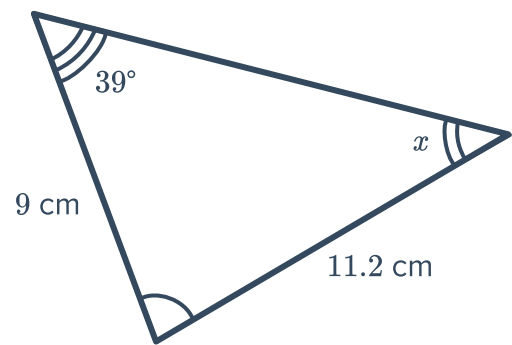
b



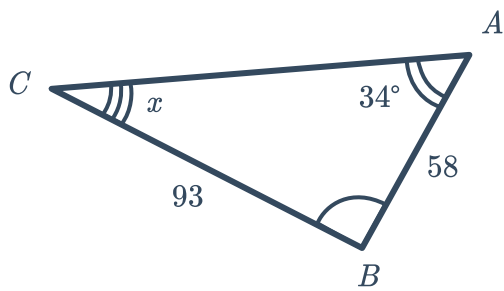
c



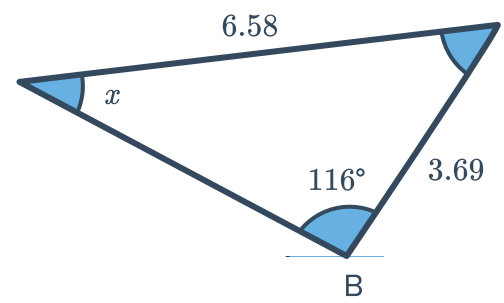
d



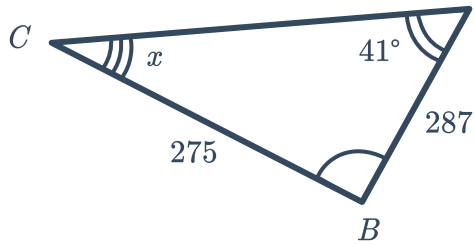
e



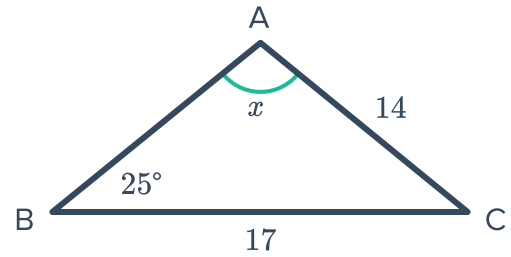
f



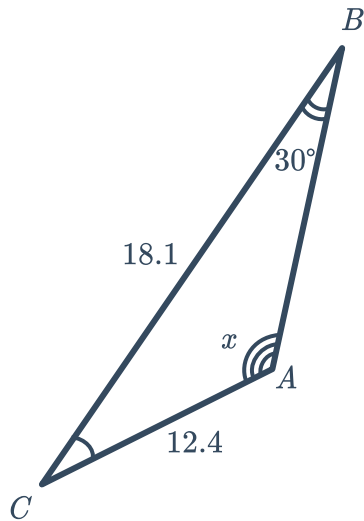
g



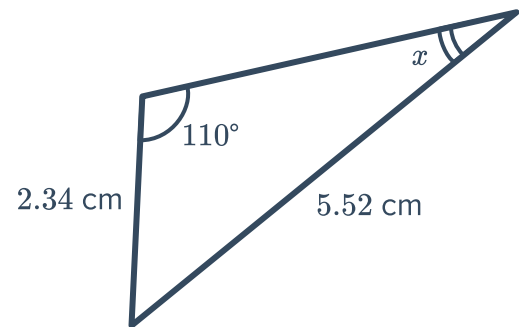
h



i



j



10 For each of the following acute angled triangles, calculate the size of $\angle B$ to the nearest degree:

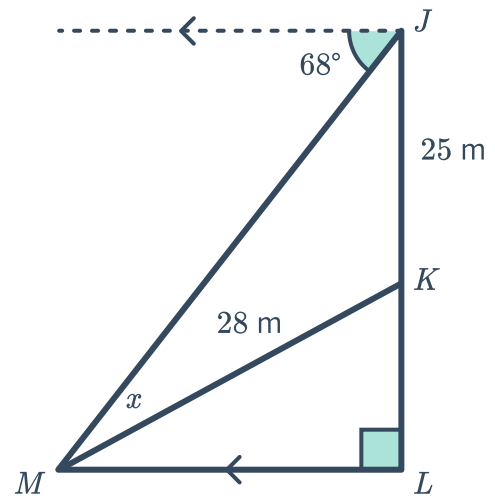
a $\triangle ABC$ where $\angle A = 57^\circ$ side $a = 156 \text{ cm}$ and side $b = 179 \text{ cm}$

b $\triangle ABC$ where $\angle A = 48^\circ$ side $a = 2.7 \text{ cm}$ and side $b = 1.9 \text{ cm}$

- 11 The angle of depression from J to M is 68° .
The length of JK is 25 m and the length of MK is 28 m as shown:

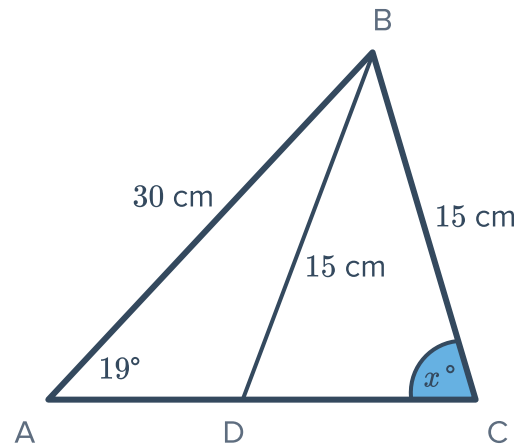
Find the following, rounding your answers to two decimal places:

- Find x , the size of $\angle JMK$.
- Find the angle of elevation from M to K .



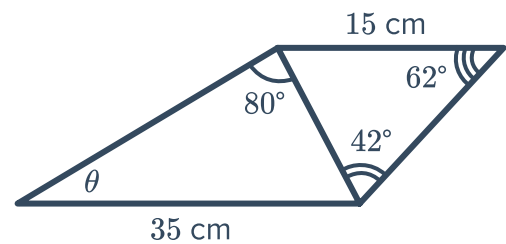
- 12 Consider $\triangle ABC$ below:

- Find x to the nearest degree.
- Find $\angle ADB$ to the nearest degree.



- 13 Consider the following diagram of a quadrilateral:

Find the value of θ , correct to two decimal places.

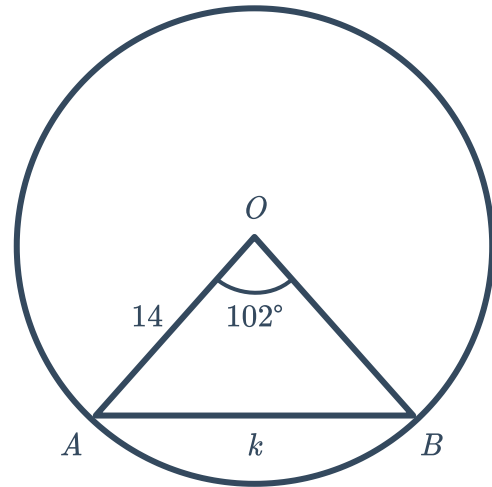




Applications

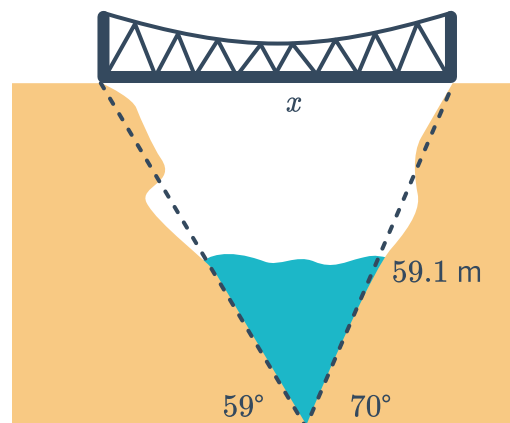
14 Consider the following diagram:

- a Find the size of $\angle OBA$.
- b Find the length of k , to two decimal places.



15 A bridge connects two towns on either side of a gorge, where one side of the gorge is inclined at 59° and the other side is inclined at 70° . The length of the steeper incline is 59.1 m.

Find x , the length of the bridge. Round your answer correct to one decimal place.



- 16 During football training, the coach marks out the perimeter of a triangular course that players need to run around. The diagram shows some measurements taken of the course, where side length $a = 14$ m:



- a Find the size of $\angle A$.
- b Find the length of side c , correct to two decimal places.
- c Find the length of side b , correct to two decimal places.
- d Each player must sprint one lap and then jog one lap around the triangle. This process is to be repeated 3 times by each player.

If Tara can run 280 m/min , and can jog at half the speed she runs, calculate the time this exercise will take her, correct to one decimal place.

